

Abhishek Nagwani

Hyderabad, IN | GMAIL | +91 9579767131 | LinkedIn | Portfolio

Mechanical Engineer | Thermal & Simulation Engineer | Structural Dynamics Engineer

Mechanical Engineer with experience in mechanical design, CFD/FEA simulations, and multidisciplinary system development. Skilled in thermofluid and structural analysis, with interests in thermal systems, structural dynamics, and simulation-driven engineering solutions.

Internships

Mechanical Design Intern, ERIC Robotics Feb 2026 - Present

- Engineered a multi-gauge railway/runway inspection robot with profilometer-based scanning targeting micron-level accuracy for detecting track and railhead irregularities; validated structural durability, terrain adaptability, and inspection reliability through CAE simulations and ground testing.

FEA Intern, Vintillix Innovations Nov 2025 - Dec 2025

- Conducted structural, thermal, and dynamic FEA in Abaqus on robotic and electronic assemblies, evaluating heat gradients, vibration response, and modal characteristics to identify critical hotspots, resonance risks, and structural weaknesses.

Production Intern, Manjira Machine Builders May 2025 - Jul 2025

- Contributed to the manufacturing of ISRO's Gaganyaan module components, interpreting design specifications and validating fabricated aerospace parts to ensure dimensional accuracy and compliance with ISRO standards.

Projects

Manufacturing and Analysis of Crew Module Components for ISRO's Gaganyaan Mission May 2025 - Jul 2025

- Interpreted technical design data provided by ISRO to assist in the precise fabrication of critical components of the crew module.

Lead Engineer, Project Garuda-1 (Autonomous 1kg Payload UAV) July 2025 - Jan 2026

- Designed and built Project Garuda-1, a custom autonomous quadcopter platform, integrating propulsion, avionics, and FPV systems.

Godawari Solar Thermal Power Plant Efficiency Enhancement May 2025 - August 2025

- Conducted thermodynamic analysis of the Godawari Solar Thermal Power Plant using TRNSYS and analytical calculations, improving thermal to electric efficiency from 15% to 19.2% by replacing Therminol VP-1 with molten salt and integrating a supercritical CO2 Brayton cycle in place of the conventional Rankine cycle.

Education

Bachelor Of Engineering in Mechanical Engineering 2022 – 2026
Chaitanya Bharathi Institute Of Technology GPA: 8.5

Publications

Design and Analysis of Crew Module for Atmospheric Re-Entry 2026
IJSRED V9I3P1

Technical Skills

Languages: Python, MYSQL , CNC Programming

CAD/CAE: SolidWorks, CATIA , Onshape, AUTOCAD , Fusion, PowerMill , Abaqus(Simulia)

SIMULATION: ANSYS, TRNSYS, System Advisor Model, MATLAB

Embedded Systems: ARDUINO UNO, Mission Planner